MCQ A · reading and checking a representation

Orthonormal coordinate system. After S67; MCQ B after S70.

For each question, only one option is correct. Justify y
looking at the solutions.

1. $d : (x;y;z) = (1;0;2) + t(2;-1;0)$ has direction vector:

A. $(1;0;2)$ B. $(2;-1;0)$

C. $(2;1;0)$ D. $(0;0;2)$

2. Which of these points belongs to d ?

A. $(3;1;2)$ B. $(1;0;0)$

C. $(2;-1;0)$ D. $(3;-1;2)$

3. A normal vector to the plane $x + 2y - 2z = 3$ is:

A. $(1;2;-2)$ B. $(1;2;3)$

C. $(3;0;0)$ D. $(-2;1;0)$

4. To check that H is proj. of M on $\mathcal{P}$, we need:

A. only $H \in \mathcal{P}$ B. only MH small

C. $H \in \mathcal{P}$ and $\overrightarrow{MH}$ normal to the plane D. that $H = M$

5. The line $A + tu$ with $u \neq 0$ is also represented by:

A. $A + t\vec{0}$ B. $A + 2u + s(3u)$

C. $tA + u$ D. the single point $A + u$

MCQ B · intersections and special cases

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Substituting a line into a plane equation gives $0 = 4$. The intersection is:

A. a line B. a point

C. the plane D. empty

7. If substitution gives $0 = 0$ for every parameter value:

A. the line lies in the plane B. there is no common point

C. there is exactly one common point D. the data are wrong

8. To find the intersection of $A + tu$ and $B + sv$, we must:

A. set $s = t$ B. replace both parameters with 0

C. solve for s, t independent D. ignore the third coordinate

9. The intersection of $x + y = 1$ and $z = 2$ is:

A. a plane B. the line $(t;1-t;2)$

C. the point $(1;0;2)$ D. empty

10. For $m = 0$, the set $mx + y = 1$ is:

A. empty B. a line

C. the whole of space D. the plane $y = 1$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Two lines whose direction vectors are not collinear necessarily intersect.

2. The planes $x + y + z = 1$ and $2x + 2y + 2z = 3$ coincide.

3. The projection of M onto a plane is unique.